

5.2 Classwork: Calculus Review (no calculator)

Four problems, one per page. Answer in the spaces provided and show all working. Total: 38 marks.

1. [Maximum mark: 8]

AA HL 2021 May P2 TZ2 Q9

(a) Write down the first three terms of the binomial expansion of $(1+t)^{-1}$ in ascending powers of t . [1]

(b) By using the Maclaurin series for $\cos x$ and the result from part (a), show that the Maclaurin series for $\sec x$ up to and including the term in x^4 is

$$1 + \frac{x^2}{2} + \frac{5x^4}{24}.$$

[4]

(c) By using the Maclaurin series for $\arctan x$ and the result from part (b), find

$$\lim_{x \rightarrow 0} \left(\frac{x \arctan 2x}{\sec x - 1} \right).$$

[3]

Nickname:

2. [Maximum mark: 16]

AA HL 2021 Nov P1 Q10

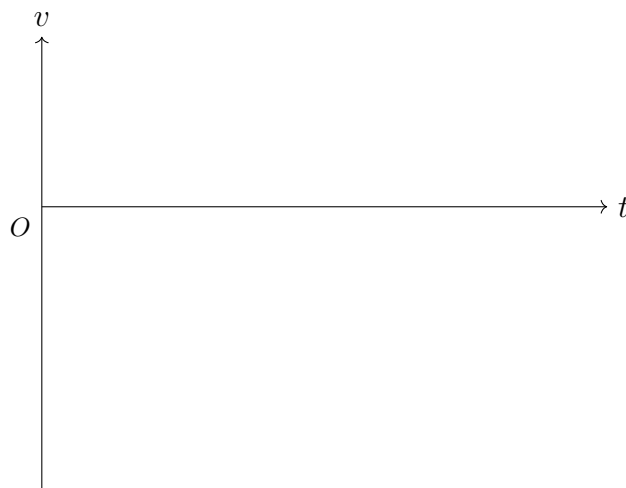
A particle P moves along the x -axis. The velocity of P is v m s⁻¹ at time t seconds, where $v(t) = 4 + 4t - 3t^2$ for $0 \leq t \leq 3$. When $t = 0$, P is at the origin O .

(a) (i) Find the value of t when P reaches its maximum velocity.

(ii) Show that the distance of P from O at this time is $\frac{88}{27}$ metres. [7]

(b) Sketch a graph of v against t on the axes below, clearly showing any points of intersection with the axes. [4]

(c) Find the total distance travelled by P . [5]



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3. [Maximum mark: 7]

AA HL 2021 May P1 TZ2 Q9

By using the substitution $u = \sin x$, find

$$\int \frac{\sin x \cos x}{\sin^2 x - \sin x - 2} dx.$$

[7]

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4. [Maximum mark: 7]

AA HL 2021 Nov P1 Q8

Solve the differential equation

$$\frac{dy}{dx} = \frac{\ln 2x}{x^2} - \frac{2y}{x}, \quad x > 0,$$

given that $y = 4$ at $x = \frac{1}{2}$.

Give your answer in the form $y = f(x)$.

[7]